

Computer Fundamentals & Hardware

A beginner-friendly starting point for understanding how computers are built and how the major hardware components work.

Recommended learning order

1. Computer basics
2. CPU / memory / storage
3. How digital hardware represents information
4. Build intuition before moving to deeper architecture

Direct YouTube resources

Source	What to use it for	Open
CrashCourse Computer Science	Computer Science playlist — start with the early episodes for computer basics and hardware concepts.	YouTube →
Ben Eater	Why build an entire computer on breadboards? — a practical look at what is inside a computer.	YouTube →
Ben Eater	Build a computer from scratch — 6502 computer series.	YouTube →

Tip: open the YouTube → buttons directly from the PDF. This roadmap is based on the supplied Hardware Learning Roadmap, with Computer Networks added as a separate course because you specifically requested it.

Prepared as a separate topic-wise learning PDF.